FSK116: Tutorial Problem Numbers / Tutoriaal Probleem Nommers

2017

2 Week: 20 February 2017 / 20 Februarie 2017

All the inidcated numbers refer to the "Problems" section in Serway; e.g. P1.9 is problem 9 from chapter 1.

Alle nommers verwys na die "Problems" afdeling in Serway; bv. P1.9 is probleem 9 uit hoofstuk 1.

P2.56 (a) $|v_p| = v_i + gt$ (b) $d = \frac{1}{2}gt^2$ (c) $v_p = |v_i - gt|$ and $d = \frac{1}{2}gt^2$
P2.63 (a) 3.00 s (b) -15.3 m/s (c) $v_{1f} = -31.4$ m/s and $v_{2f} = -34.8$ m/s
P2.72 (a) 41.0 s (b) 1.73 km (c) -184 m/s

P3.11 (a) 5.2 m at 60° (b) 3.0 m at 330° (c) 3.0 m at 150° (d) 5.2 m at 300°
P3.12 (a) graphical (b) $\vec{R}_1 = \vec{R}_2 = \vec{R}_3$
P3.15 47.2 units at 58.0° above the negative x-axis
P3.20 (a) 5.00 blocks at 53.1° North of East (b) 13.0 blocks
P3.23 (a) $2\hat{\imath} - 6\hat{\jmath}$ (b) $4\hat{\imath} + 2\hat{\jmath}$ (c) 6.32 (d) 4.47 (e) $\theta_{|A+B|} = 288°$ and $\theta_{|A-B|} = 26.6°$
P3.45 1.43×10^4 m at 32.2° above the horizontal
P3.51 (a) $(-130\text{ m})\hat{\imath} - (202\text{ m})\hat{\jmath}$ or 240 m at 57.2° below the horizontal
P3.60 $\theta = 2\tan^{-1}(\frac{1}{n})$ Hint: Draw $\vec{A} + \vec{B}$ and $\vec{A} - \vec{B}$ graphically. You will need to use the law of cosines and you may need the identity $1 - \cos\theta = 2\sin^2(\theta/2)$.
P3.61 (a) $|\vec{A} + \vec{B}| = [10000 - (9600)\sin\theta]^{1/2}$ cm (b) $\theta = 270°$, 140.0 cm (c) $\theta = 90°$, 20.0 cm (d) explanation
P3.62 $\Delta\vec{r} = (0.456\text{ m})\hat{\imath} - (0.708\text{ m})\hat{\jmath}$
P3.63 (a) $\frac{d\vec{r}}{dt} = -(2.00\text{ m/s})\hat{k}$ (b) explanation.